

2

Understanding and Analysing Spot Prices

2.1 Introduction

In this chapter we describe how the Black-Scholes geometric Brownian motion model, introduced in chapter one, can be generalised to a more realistic model for spot energy prices. Other useful discussions of modelling energy prices can be found in Humphreys and Shimko (1997), Pilipovic (1997) and Eydeland and Geman (1998). We discuss mean reversion – the tendency of spot prices to move back towards their long term level, stochastic volatility – the unpredictability of spot price volatility and jumps – sudden large changes in the spot price. The Monte Carlo simulation of these processes is also described, both as an aid to understanding and to begin to develop the numerical techniques needed to price energy derivatives. Simple yet robust estimation methods for the key parameters of the mean reversion and jump models are presented with illustrative numerical examples. The estimation of volatility models is discussed in depth in chapter 3.

We begin discussing the various extensions to the BSM model by assuming constant parameters for ease of explanation. In reality the volatility, mean reversion rate, long term level and jump behaviour will at the very least vary through time with reasonably predictable “seasonal” patterns. Extending the models to handle these seasonal patterns is generally straightforward and we discuss this in more detail in the penultimate section of this chapter. In particular, time varying parameters are very easy to incorporate into the Monte Carlo simulations which we discuss in this chapter and chapters 7 and 8.

Another important energy derivative pricing issue, introduced in chapter 1, is the skew and smile observed in Black-Scholes implied volatilities which is directly related to skews and fat-tails in the distributions of energy price returns. Fat-tails can be accounted for by incorporating random volatility and more importantly jumps into the models which we discuss in detail in this chapter. Skews are more difficult to incorporate but we mention some approaches in the sections on modelling volatility and jumps. Finally, an issue that occurs in electricity markets is that of negative prices (see section 3.1 for a discussion of this). The models that we discuss are generally designed to exclude the possibility of such prices. If the frequency and range of the negative prices is small relative to the positive prices then they can be excluded without affecting the results discussed in this book. Alternatively, a lower bound can usually be determined for the prices and the actual

prices can then be shifted so as to be positive. Although the analytical results we discuss will no longer hold, the numerical methods can still be used to price derivatives.

2.2 Mean Reversion

It has been well documented that an important property of energy spot prices is mean-reversion (Gibson and Schwartz (1990), Brennan (1991), Cortazar and Schwartz (1994), Bessembinder, Coughenour, Seguin and Smoller (1995), Ross (1995)). Mean reversion can be understood by looking at a simple model of a mean reverting spot price (Schwartz (1997)) represented by the following equation;

$$dS = \alpha(\mu - \ln S)Sdt + \sigma Sdz \quad (2.1)$$

In this model the spot price mean reverts to the long-term level $\bar{S} = e^\mu$ at a speed given by the mean reversion rate, α which is taken to be strictly positive. If the spot price is above the long-term level $\bar{S}$ then the drift of the spot price will be negative and the price will tend to revert back towards the long-term level. Similarly, if the spot price is below the long-term level then the drift will be positive and the price will tend to move back towards $\bar{S}$. Note that at any point in time the spot price will not necessarily move back towards the long term level as the random change in the spot price may be of the opposite sign and greater in magnitude than the drift component. Figure 2.1 shows the Henry Hub natural gas spot price for the period from 1 April 1999 to 31 March 2000.

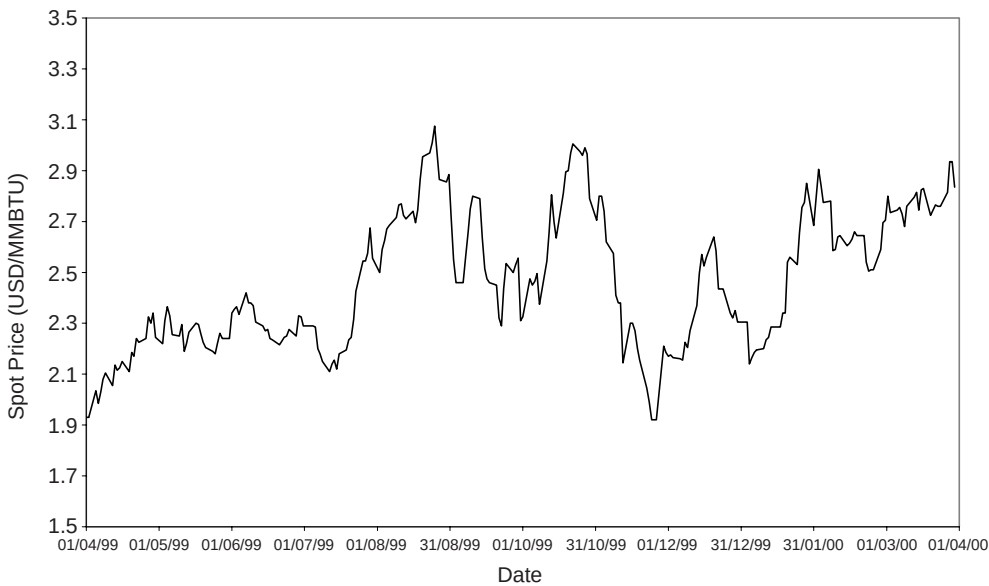


FIGURE 2.1 Henry Hub Natural Gas Spot Price for the Period from 1 April 1999 to 31 March 2000 (USD/MMBTU)

The mean reversion rate of the spot gas price over this period is approximately eleven (estimation of the mean reversion rate is discussed in section 2.9). This implies that the spot price tends to be pulled back to its long-term level over a period of roughly a month (this is related to the half-life which is discussed in detail in section 2.4). In reality the spot price does not mean revert to a constant long term level, and we show in chapters 6, 7 and 8 how information on the level to which the spot price mean reverts is contained in the forward curve prices and volatilities.

2.3 Simulating Mean Reversion

The simple model of a mean reverting spot price represented by equation (2.1) can be written in terms of the natural logarithm of the spot price, $x = \ln S$, as follows

$$dx = [\alpha(\mu - x) - \frac{1}{2}\sigma^2]dt + \sigma dz \quad (2.2)$$

Equation (2.2) can be discretised as

$$\Delta x_i = [\alpha(\mu - x_i) - \frac{1}{2}\sigma^2]\Delta t + \sigma\sqrt{\Delta t}\varepsilon_i \quad (2.3)$$

Note that in this case, unlike for GBM (see chapter one), since the drift term depends on the variable x the discretisation is only correct in the limit of the time step tending to zero. We must therefore choose time steps which are small relative to the speed of mean reversion - that is a small fraction of the half-life. In order to obtain a simulated path of the spot price we proceed in exactly the same way as for GBM in chapter one. Firstly, we choose or estimate the parameters α , μ , σ and Δt ¹. Then we repeatedly generate normally distributed random numbers ε_i and from these calculate each new value of x , and thus the new spot price at each time step. Figures 2.2 and 2.3 illustrate a simulated GBM path and mean reverting path for the energy spot price respectively. In both figures exactly the same random shocks have been applied in the spot price simulations.

We have set the mean reversion rate in figure 2.3 relatively high, $\alpha = 100$, in order that the mean reverting behaviour can be easily seen relative to figure 2.2. Comparing figures 2.2 and 2.3, it should be clear that the same random shocks have occurred to the spot price. However, the range of the spot price in figure 2.3 over time is reduced by the effect of the mean reversion continuously driving the spot price towards the long-term mean of 100.

2.4 The Half-Life of a Mean Reverting Process

A key property of a mean reverting process is the half-life. This is the time taken for the price to revert half way back to its long-term level from its current level if no more

¹ We discuss estimation of the mean reversion rate in the following section. In chapter 6, 7 and 8 we show the parameter μ can be obtained as a time dependent function from the forward price and volatility curve. Estimation of volatility is discussed in chapter 3.

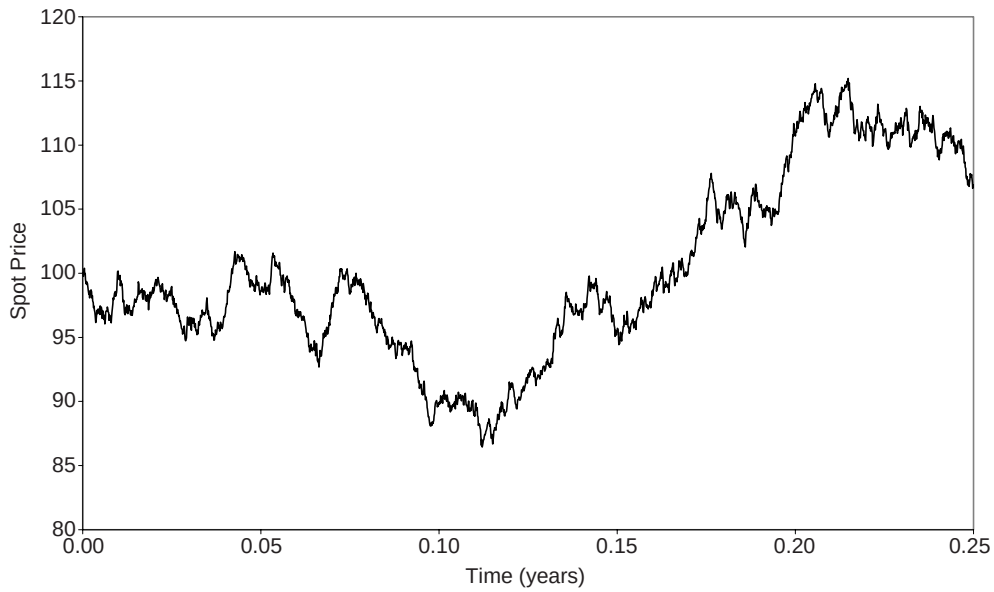


FIGURE 2.2 Simulation of a Geometric Brownian Motion Spot Price Path
 $S = 100, r - \delta = 0.05, \sigma = 0.30, \Delta t = 1/(365 \times 24)$

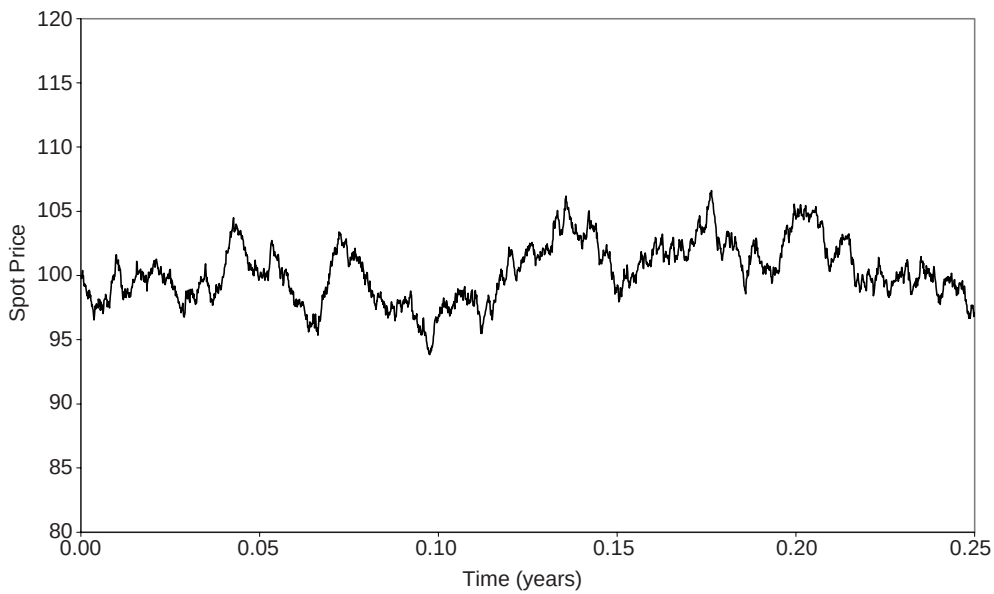


FIGURE 2.3 Simulation of a Mean Reverting Path for the Spot Energy Price
 $S = 100, \alpha = 100.0, \bar{S} = 100, \sigma = 0.30, \Delta t = 1/(365 \times 24)$

random shocks arrive. Ignoring the randomness allows us to focus on the mean reverting behaviour alone. Consider the simple mean reverting process, in terms of $x = \ln S$, reverting to a level of $\bar{x}$. In the absence of randomness we have

$$dx = \alpha(\bar{x} - x)dt \quad (2.4)$$

Equation (2.4) can be solved by integrating from x_0 at time 0 to x_t at time t to obtain

$$(x_t - \bar{x}) = (x_0 - \bar{x})e^{-\alpha t} \quad (2.5)$$

To obtain the half-life, which we denote by $t_{1/2}$, at which the distance of x from its long term level is half its initial distance we have

$$\frac{1}{2}(x_0 - \bar{x}) = (x_0 - \bar{x})e^{-\alpha t_{1/2}} \quad (2.6)$$

Solving for $t_{1/2}$ we obtain

$$t_{1/2} = \ln(2)/\alpha \quad (2.7)$$

Table 2.1 gives the half-life for a range of values of the mean reversion rate.

It is important to realise that these times are averages. Only on average over a long time period, will shocks to the spot price take the time given by the half-life to decay to half their deviation from the long term level.

2.5 Stochastic Volatility

Even a cursory examination of most energy prices suggests that the volatility of the price is changing through time. Figure 2.4 shows the Brent crude oil spot price for the period from 1 April 1999 to 31 March 2000.

The volatility of the spot oil price over the first part of the data is clearly less than that over the second half of the data. Many models have been proposed for the behaviour of volatility (for example – Hull and White (1987), Johnson and Shanno (1987), Wiggins (1987), Hull and White (1988), Madan and Senata (1990), Stein and Stein (1991), Heston (1993), Derman and Kani (1994), Dupire (1994), Duan (1995), Bates (1996), and Scott (1997), Ritchken and Trevor (1999)). Here we focus on two particular types of model. The first model we discuss was introduced independently by Derman and Kani (1994)

TABLE 2.1 Mean Reversion Rates and the Corresponding Half-Lives

α	$t_{1/2}$
1	8 months
10	25 days
100	2.53 days
1000	6 hours

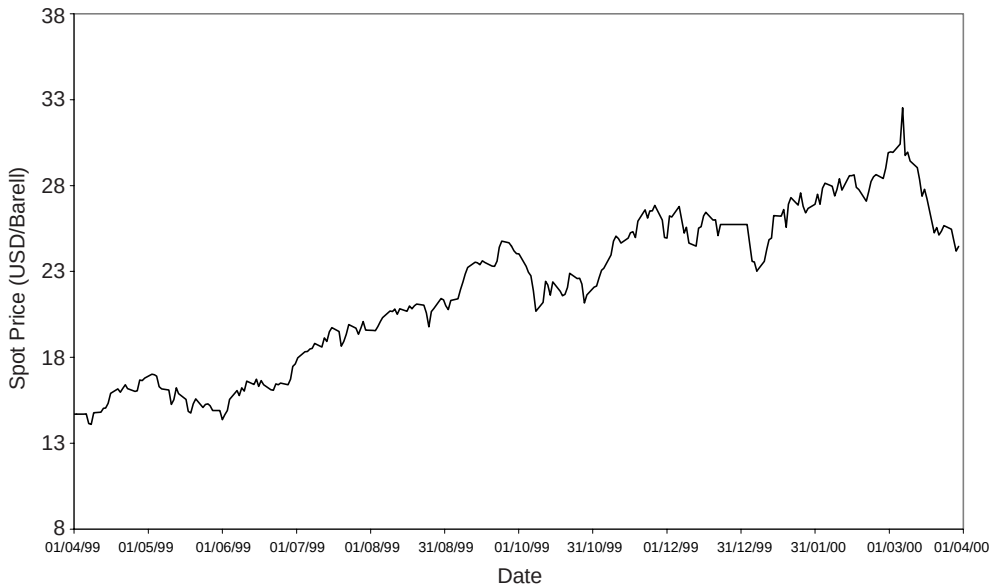


FIGURE 2.4 Brent Crude Oil Spot Price for the Period from 1 April 1999 to 31 March 2000 (USD/Barrel)

and Dupire (1994). The idea behind this model was that it could be fitted to or implied from the market prices of standard European options via a trinomial tree, and it is therefore often referred to as the implied tree method². The second model was first proposed by Hull and White (1988) and subsequently used and extended by Heston (1993), Bates (1996) and Scott (1997).

The modelling approach of Derman *et al* and Dupire can be represented by the following process for the spot price;

$$dS = \mu S dt + \sigma(t, S) dz \quad (2.8)$$

In equation (2.8) the volatility is now a general function of the spot price and time. If this function is specified to depend on a non-linear function of the spot price then the model allows us to obtain spot price returns distributions with skews. A simple example would be;

$$\sigma(t, S) = \beta_1 S^{\beta_2} \quad (2.9)$$

where β_1 and $\beta_2 > 1$ are constants – this is referred to as the constant elasticity of variance model (see Cox and Ross (1976)). The general model given by equation (2.8)

² This modelling approach is also discussed in chapter 7. For a detailed discussion of the implementation of implied trinomial trees see Clewlow and Strickland (1998), chapter 5.

can be simulated in the same way as GBM except for one small complication. Transforming to the natural logarithm of the spot price no longer leads to a constant volatility for the process. In some simple cases, such as equation (2.9), it is possible to find a transformation which leads to a variable with a constant volatility (see Clewlow and Strickland (1998), chapter 8 for a discussion of this in the context of interest rate models). If a transformation is not possible then small time steps must be used to simulate the process accurately.

The second type of model became popular because of its realistic properties and computational tractability. It is described by the following processes for the spot price and the spot price return variance $V = \sigma^2$;

$$dS = \mu S dt + \sigma S dz \quad (2.10)$$

$$dV = a(\bar{V} - V)dt + \xi\sqrt{V}dw \quad (2.11)$$

Equation (2.10) is the GBM model but with a volatility, σ , which is no longer constant but which changes randomly. The behaviour of the volatility is determined by equation (2.11) which specifies the process followed by the variance – the square of the volatility. The variance mean reverts to a long-term level $\bar{V}$ at a rate given by a . The absolute volatility of the variance is $\xi\sqrt{V}$ which is proportional to the square root of the variance i.e. the volatility of the spot price. The source of randomness in the variance, dw , is different from the dz driving the spot price although it may be correlated with correlation coefficient ρ . A more detailed discussion of stochastic volatility and estimation of the parameters of the volatility process is presented in chapter 3.

2.6 Simulating Stochastic Volatility

The stochastic volatility model described by equations (2.10) and (2.11) can be discretised in the following way³;

$$\Delta x_i = \left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_{1,i} \quad (2.12)$$

$$\Delta V_i = a(\bar{V} - V_i)\Delta t + \xi\sqrt{V_i}\Delta t\left(\rho\varepsilon_{1,i} + \sqrt{1 - \rho^2}\varepsilon_{2,i}\right) \quad (2.13)$$

where ε_1 and ε_2 are independent standard normal random variables. These are combined through the correlation coefficient ρ to obtain a random shock to the variance, which is correlated with the random shock to the spot price (simulating correlated processes is discussed more fully in chapters 7 and 8). The drift and volatility terms in both equations (2.12) and (2.13) depend on the variance and so the discretisation is only correct in the limit of the time step tending to zero. We must therefore choose time steps which are small relative to the speed of mean reversion and the volatilities. The simulation of a spot price path now requires the joint simulation of both the spot price and the variance of the

³ The spot price process is discretised in terms of its natural logarithm x .

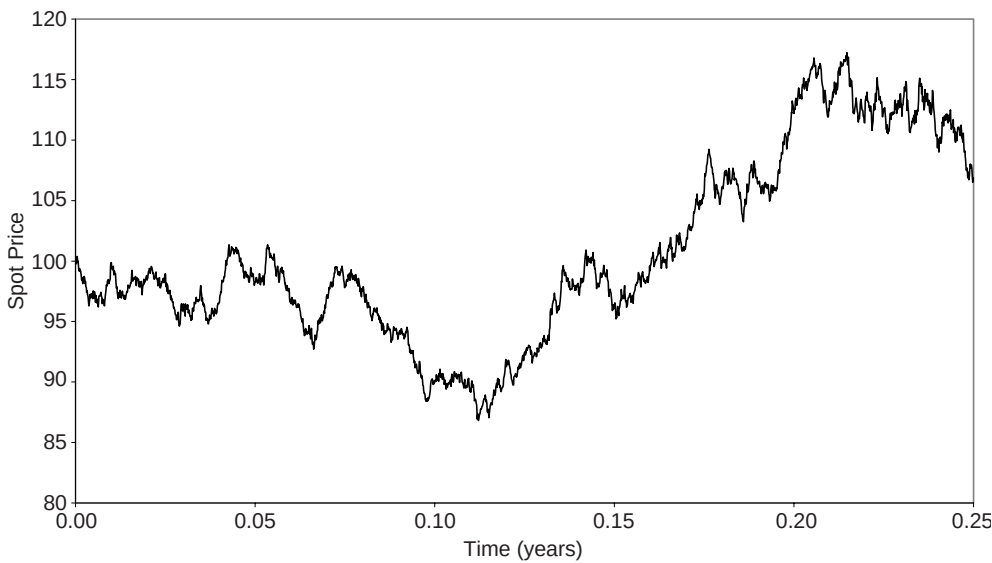


FIGURE 2.5 (a) A Simulated Stochastic Volatility Path for the Spot Energy Price
 $S = 100, \mu = 0.05, \sigma = 0.30, a = 10, \bar{V} = 0.09, \xi = 0.5, \rho = 0.0, \Delta t = 1/(365 \times 24)$

spot price. We first obtain or choose values for $\mu, \sigma, a, \xi, \rho$, and Δt . Each time step of the simulation then involves generating two independent normally distributed random numbers ε_1 and ε_2 , from which the variance and the spot price at the end of the time step can then be calculated. Repeating this process yields a simulated path for both the spot price and its variance. Figures 2.5 (a) and 2.5 (b) illustrate simulated paths for the spot price and the variance of the spot price in which the random shocks to the spot price are the same as in figure 2.2 and 2.3.

From a comparison of figures 2.2 and 2.5 it can be seen that although the random movements are in the same directions, the movements are larger in the later part of figure 2.5 where the variance has increased.

2.7 Jumps in the Spot Price

Energy prices often exhibit sudden, unexpected and discontinuous changes. Particularly good examples of this occur in electricity markets. Figure 2.6 is a plot of the Australian New South Wales (NSW) electricity pool price for the period from 13 December 1998 to 8 March 1999.

Figure 2.6 shows that the presence of jumps is a significant component of the behaviour of the NSW electricity pool price⁴. However, it should also be noted that the price does not stay at the level to which it jumps, but, after a jump, rapidly reverts to

⁴ See also Joy (1998) for a discussion of the application of jumps to the Australian electricity markets.

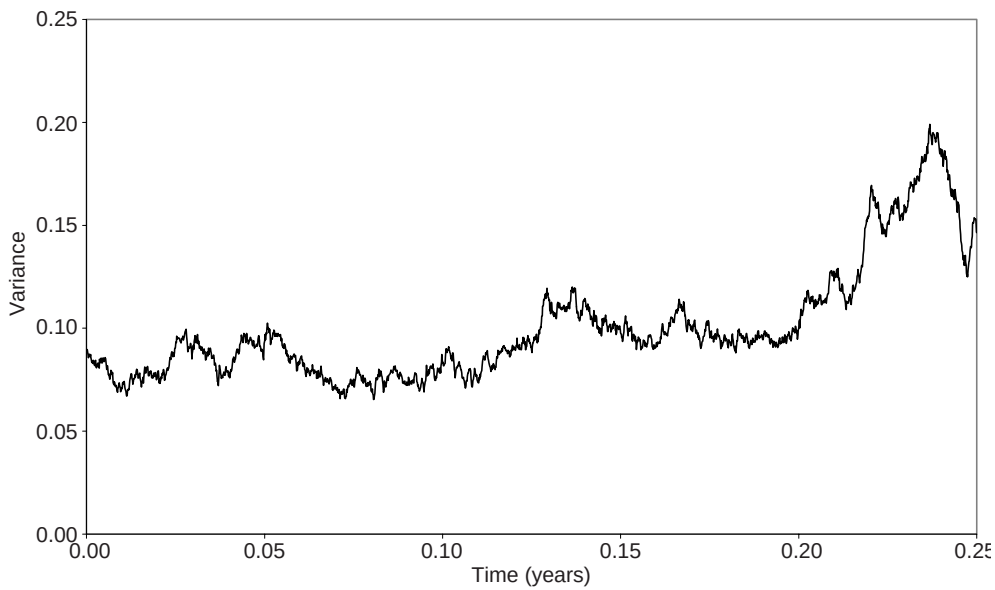


FIGURE 2.5 (b) A Simulated Stochastic Volatility Path for the Spot Energy Price
 $S = 100, \mu = 0.05, \sigma = 0.30, a = 10, V = 0.09, \xi = 0.5, \rho = 0.0, \Delta t = 1/(365 \times 24)$

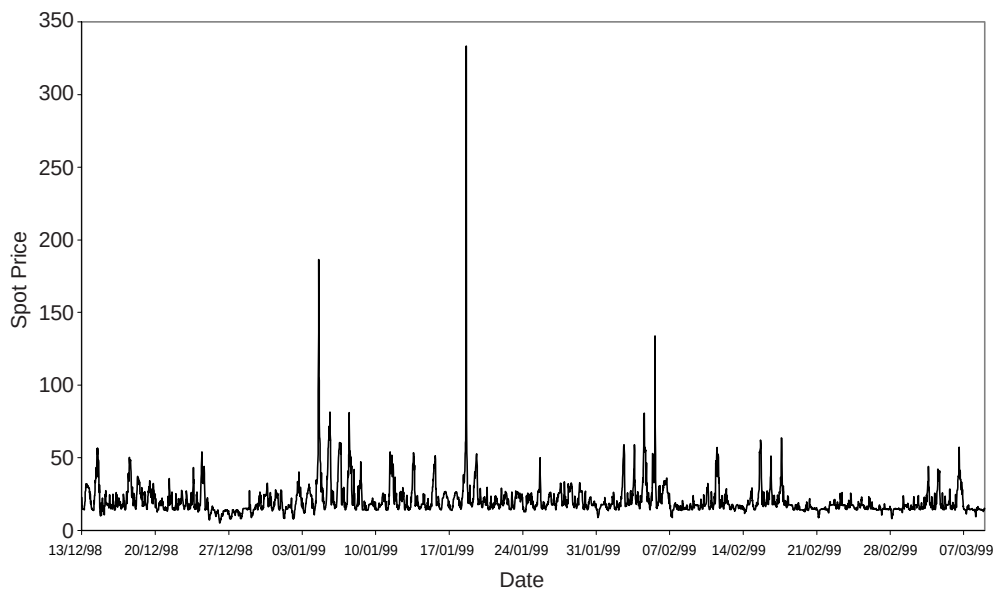


FIGURE 2.6 Australian New South Wales Electricity Pool Prices for the Period from 13 December 1998 to 8 March 1999

its long-term level. This is an important aspect of the electricity price, which we will deal with in the following section. This type of behaviour where the price exhibits sudden, large changes can be modelled by using jump processes. A simple and realistic model for a spot price, which is identical to the Black-Scholes model except for the addition of a jump process, is the jump-diffusion model introduced by Merton (1976)⁵. This model is described by the following SDE⁶;

$$dS = \mu S dt + \sigma S dz + \kappa S dq \quad (2.14)$$

The annualised frequency of jumps is given by ϕ , the average number of jumps per year⁷. The proportional jump size is κ which is random and is determined by the natural logarithm of the proportional jumps being normally distributed;

$$\ln(1 + \kappa) \sim N(\ln(1 + \bar{\kappa}) - \frac{1}{2}\gamma^2, \gamma^2) \quad (2.15)$$

where $\bar{\kappa}$ is the mean jump size and γ is the standard deviation of the proportional jump size which we call the jump volatility. The jump process (dq) is a discrete time process – jumps do not occur continuously but at specific instants of time. Therefore, for typical jump frequencies, most of the time $dq = 0$ and only takes the value 1 when a randomly timed jump occurs. When no jump is occurring the spot price behaviour is identical to GBM and is only different when a jump occurs – this is illustrated in the following section on simulation. For example, if we imagine the jump frequency becoming very small, so that the chances of a jump occurring are close to zero, then we would get a GBM spot price. Similarly, if the jump volatility were very small, so that even if jumps were very frequent their size would be insignificant, then this would again result in a GBM process.

The proportional jumps (or equivalently jump returns) in equation (2.14) are normally distributed and therefore symmetrical. That is the number of positive and negative jumps and the range of sizes of the proportional jumps will be equal on average. In reality the distribution of jump return sizes of energy spot prices is positively skewed. A simple way to incorporate this property into equation (2.14) is to have the proportional jumps drawn from a normal distribution but with different jump volatilities for the positive and negative jumps. Another simple alternative would be to have the proportional jumps drawn from a negatively shifted lognormal distribution – this would give a lower limit on the negative jump returns. These extensions are straightforward to incorporate into Monte Carlo simulation but lead to the loss of the analytical tractability of the Merton (1976) model.

2.8 Simulating Jumps in the Spot Price

The jump-diffusion model described by equation (2.14) can be discretised as follows;

$$\Delta x_i = (r - \phi \bar{\kappa} - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_{1i} + (\bar{\kappa} + \gamma\varepsilon_{2i})(u_i > \phi\Delta t) \quad (2.16)$$

⁵ See Merton (1990) for a general discussion of jump processes applied to financial markets.

⁶ The risk-neutral process is obtained by setting $\mu = (r - \phi \bar{\kappa})$

⁷ Technically, ϕ is defined by $\text{prob}(dq = 1) = \phi dt$

where ε_1 and ε_2 are independent standard normal random variables and u is a uniform (0,1) random sample. The term $(u_i < \phi \Delta t)$ is taken to be one if the condition is true and zero otherwise - this generates jumps randomly at the correct average frequency. When a jump occurs, its size is the mean jump size plus a normally distributed random amount with standard deviation γ determined by ε_2 . In order that the frequency of jumps is correctly simulated, the time step Δt must be small relative to the jump frequency such that $\phi \Delta t \ll 1$. Once the parameters ϕ , $\bar{\kappa}$, σ , γ and Δt have been chosen, at each time step, we first generate a uniform random number, u , in order to determine if a jump should occur. If a jump is to occur we generate the normally distributed random number ε_2 which determines the size of the jump. Finally, we generate the normally distributed random number ε_1 which determines the usual GBM random change in the spot price. Figure 2.7 illustrates a simulated path for the spot price under the jump-diffusion model in which the random shocks to the spot price are the same as in figures 2.2, 2.3 and 2.5. The jumps are indicated by the bold lines with diamonds at each end.

The behaviour of the simulated spot price does not appear to be a good model for the spot electricity price, partly because the jumps are not sufficiently large but also because once a jump has occurred the spot price stays at the new level and does not mean revert. This problem can be corrected by combining mean reversion and jumps into the same model. This can be represented by the following SDE;

$$dS = \alpha(\mu - \ln S)Sdt + \sigma Sdz + \kappa Sdq \quad (2.17)$$

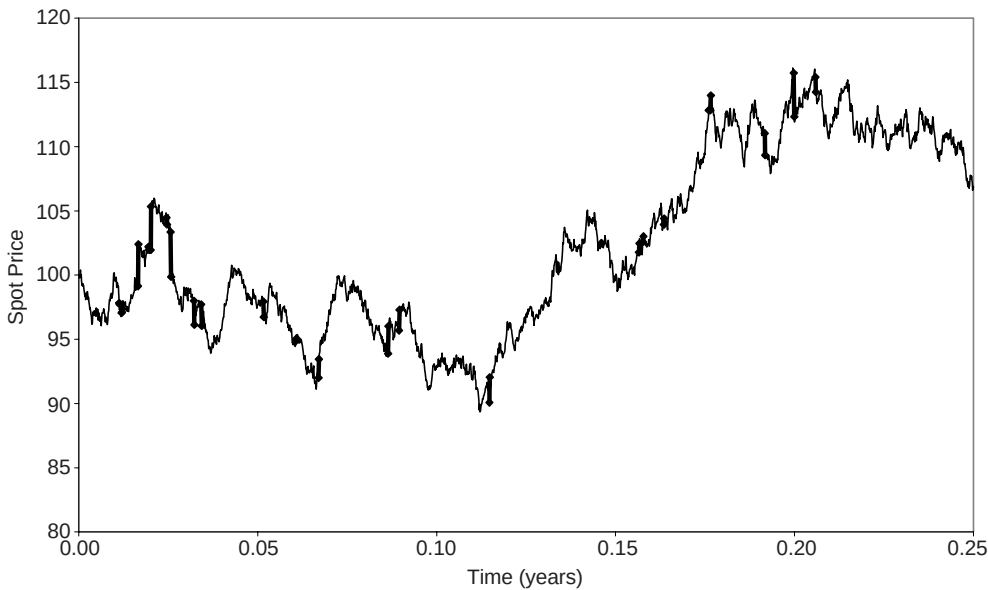


FIGURE 2.7 A Simulated Jump-Diffusion Path for the Spot Energy Price
 $S = 100, r = 0.05, \sigma = 0.30, \bar{\kappa} = 0, \phi = 100, \gamma = 0.02, \Delta t = 1/(365 \times 24)$

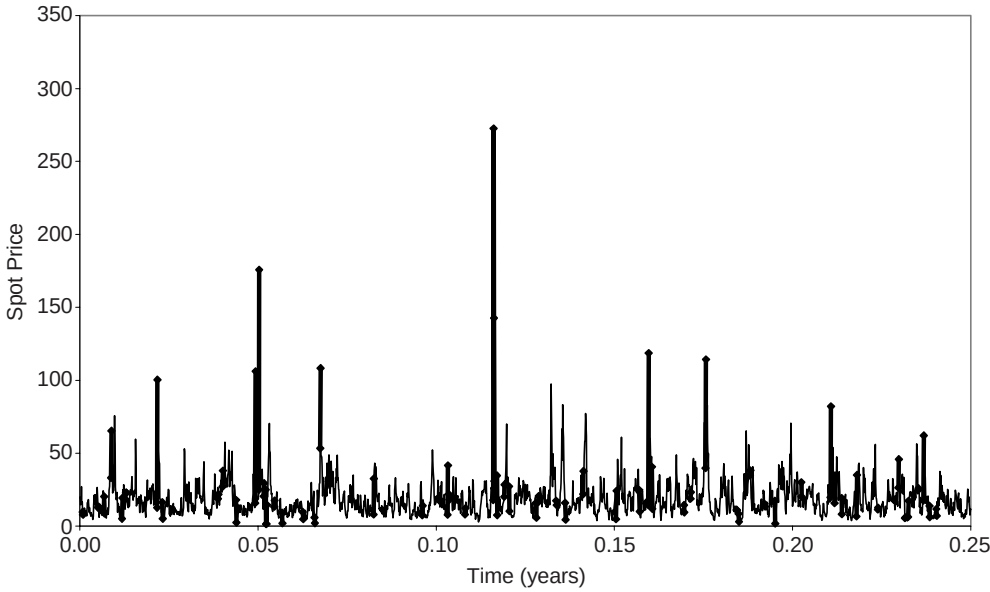


FIGURE 2.8 A Simulated Mean Reverting Jump-Diffusion Path for an Electricity Spot Energy Price $S = 20, \alpha = 2000, \bar{S} = 20, \sigma = 30.0, \bar{\kappa} = 0, \phi = 250, \gamma = 1.2, \Delta t = 1/(365 \times 24)$

Figure 2.8 shows a simulated path for the model represented by equation (2.17) with very strong mean reversion and a large jump volatility.

The behaviour of the spot price in figure 2.8 resembles much more closely that of the NSW electricity price in figure 2.6. It can be seen that the diffusion volatility, jump volatility and mean reversion of the NSW electricity price must be extremely high – we give examples of estimating these parameters in the following sections.

2.9 Estimation of the Mean Reversion Rate

The mean reversion rate of the spot energy price can be estimated relatively simply and robustly via linear regression. Consider the simple mean reverting process for the natural logarithm of the energy spot price⁸;

$$dx = \alpha(\bar{x} - x)dt + \sigma dz \quad (2.18)$$

This can be discretised as follows;

$$\Delta x_t = \alpha_0 + \alpha_1 x_t + \sigma \varepsilon_t \quad (2.19)$$

where $\alpha_0 = \alpha\bar{x}\Delta t$ and $\alpha_1 = -\alpha\Delta t$. This implies that observations of the spot price through time can be considered as observations of the linear relationship between Δx_t

⁸ This is essentially the same as equation (2.2) but with the $-1/2 \sigma^2$ incorporated into $\bar{x}$.

and x_t in the presence of noise (represented by $\sigma\varepsilon_t$). Therefore, if we regress observations of Δx_t against x_t we can obtain estimates of $\alpha_0 = \alpha\bar{x}\Delta t$ and $\alpha_1 = -\alpha\Delta t$ as the estimates of the intercept and slope of the linear relationship. Since we know the time interval between observations Δt we can obtain estimates of α and $\bar{x}$. In the following section we present a specific example using Henry Hub natural gas prices.

Example : Estimation of the Mean Reversion Rate of Natural Gas Spot Prices

The data we use in this example consists of the Henry Hub natural gas spot price for the period from 1 April 1999 to 31 March 2000. This gives 260 data items with $\Delta t = 0.003846$. Figure 2.9 illustrates the data in terms of changes Δx plotted against x and also the linear regression estimated straight-line fit to the data.

The estimates from the linear regression are (standard errors in parentheses):

$$\alpha_0 = 0.0398(0.0152)$$

$$\alpha_1 = 0.0428(0.0168)$$

which implies

$$\bar{x} = 2.5$$

$$\alpha = 11.1(4.4)$$

This value of α and standard error implies a half-life for price shocks of between 16 and 38 days.

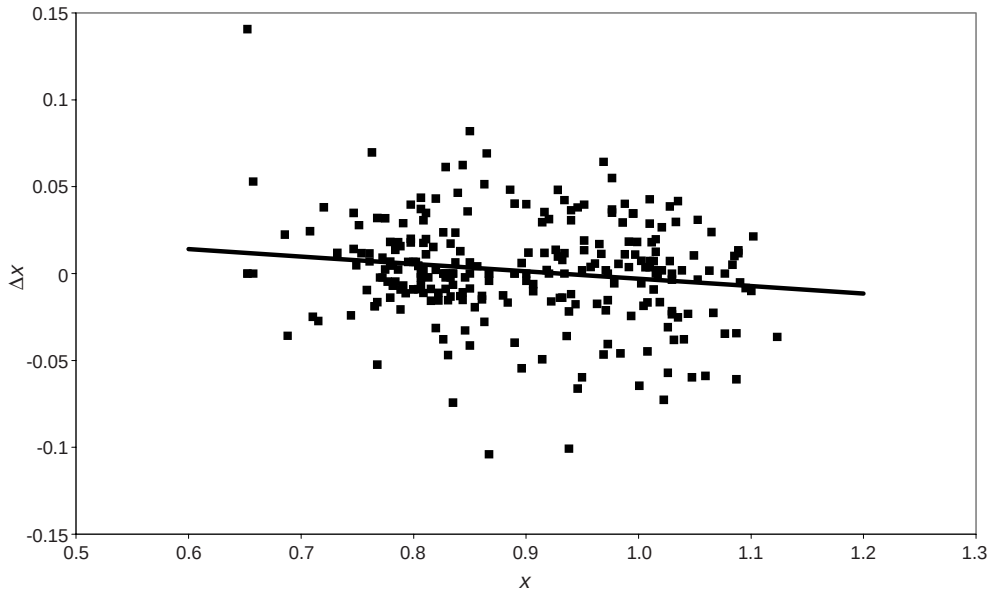


FIGURE 2.9 Linear Regression Estimation of the Mean Reversion Rate of Henry Hub Natural Gas 1 April 1999 to 31 March 2000 (USD/MMBTU)

2.10 Intuitive Estimation of Jump Process Parameters

The estimation of jump parameters for energy prices is complicated by the fact that the jumps can only be observed as part of a time series of prices which includes the “normal” non-jump behaviour of the price. Typically, we will not have information on the exact time the jumps occur. However, it is clear in figure 2.6 that the very large price “spikes” should be attributed to jumps because the probability of the normal Brownian motion type behaviour generating these large price changes is virtually zero. This observation provides a clue to how we might estimate the jump parameters. Consider figure 2.10 which shows the Australian New South Wales electricity pool prices returns (i.e. changes in the natural logarithm of the price shown in figure 2.8) every half-hour from the 13 December 1998 to the 8 March 1999.

If we assume that jumps are relatively infrequent and not too large then we can get an estimate of the diffusion volatility in the usual way by calculating the sample standard deviation of returns. Based on this estimate of the volatility we can then look for actual returns that were larger than we would expect (for a chosen probability) in the absence of jumps and identify these extreme returns as jumps. Given that we have identified some returns as jumps we should recompute the estimate of the diffusion volatility by recalculating the sample standard deviation of returns with the jump returns excluded. This will give us a lower estimate of the diffusion volatility. Consequently, using this new estimate of the diffusion volatility, we can look for further returns which exceed the chosen

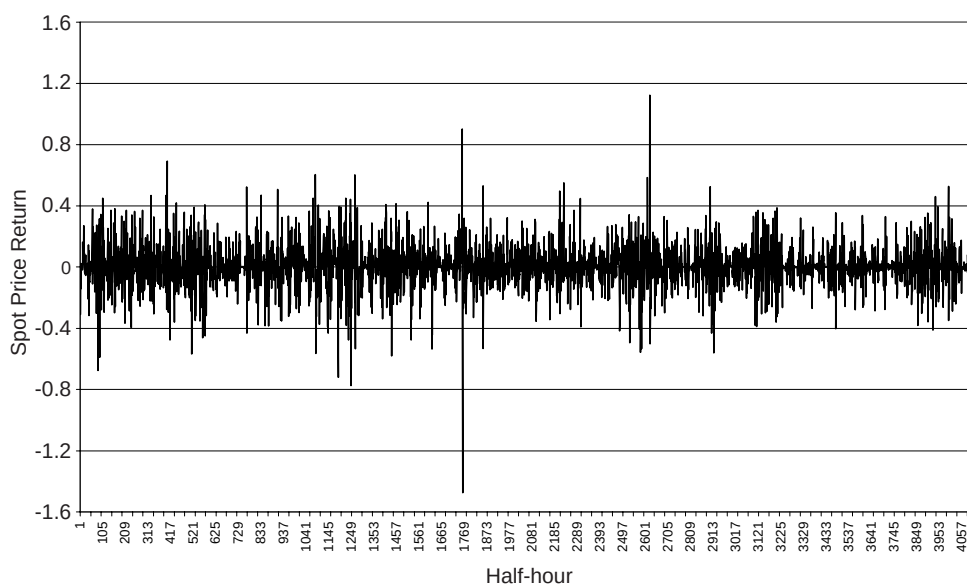


FIGURE 2.10 Australian New South Wales Pool Price Returns for 13 December 1998 to 8 March 1999

limit. This process can be repeated until the estimates converge and no new jumps are identified (typically well within ten iterations) - we call this approach a Recursive Filter.

Example : Recursive Filter Estimation of the Jump-Diffusion Parameters

The data we use in this example is the same as in figure 2.10, i.e. Australian NSW electricity pool prices every half-hour from the 13 December 1998 to the 8 March 1999. The data set therefore consists of 4128 price observations giving 4127 returns with $\Delta t = 5.70776 \times 10^{-5}$. The sample standard deviation of the half-hourly returns is 0.1353. The probability of returns greater than $3 \times 0.1353 = 0.4061$ is less than 0.003 therefore we begin by identifying returns larger in absolute value than 0.4061 as jumps. We find that there are 57 returns that exceed this limit. The number of jumps divided by the total time period over which they occur (in this case 0.2329 years) gives an estimate of the jump frequency. The mean and standard deviation of the jump returns gives us estimates of the mean jump size and the jump volatility. Thus we have the following relationships:

$$\phi = \text{number of jump returns} / \text{time period of data} \quad (2.20)$$

$$\bar{\kappa} = \text{average of jump returns} \quad (2.21)$$

$$\gamma = \text{standard deviation of jump returns} \quad (2.22)$$

Table 2.2 gives the results of repeating this process for the NSW electricity data until the estimates converge.

Note that the mean jump size is usually very difficult to estimate robustly and should therefore usually be set to zero. In the next section we describe a formal statistical method for estimating jump process parameters which has somewhat different properties.

TABLE 2.2 Iterative Estimation of the Jump-Diffusion Parameters of NSW Electricity

Iteration	SD	Jumps	ϕ	$\bar{\kappa}$	γ
1	0.1183	57	241.9	0.000149	0.5718
2	0.1121	102	432.9	-0.000107	0.4957
3	0.1088	129	547.5	-0.000197	0.4683
4	0.1076	139	589.9	-0.000696	0.4595
5	0.1065	149	632.4	-0.001022	0.4513
6	0.1058	156	662.1	-0.001105	0.4462
7	0.1050	163	691.8	-0.001348	0.4411
8	0.1043	170	721.5	-0.001271	0.4368
9	0.1041	172	730.0	-0.001430	0.4354
10	0.1041	172	730.0	-0.001430	0.4354

SD is the standard deviation of the returns after removing the jump returns,
Jumps is the number of jump returns identified.

2.11 Maximum Likelihood Estimation of Jump Parameters

Maximum likelihood is a popular method of estimating the parameters of stochastic process when the probability density for the stochastic variable can be written down analytically. It was used successfully by Ball and Torous (1983, 1985) to estimate jump parameters for NYSE stock prices when earlier work by Beckers (1981) using the method of moments often gave negative variance estimates. As we will return to maximum likelihood estimation in section 3.4.3 we will only give an overview of the technique in this section. The idea behind maximum likelihood is based on the observation that for a given choice of parameter values, Θ , we can compute the probability or likelihood, $L(\Delta x; \Theta)$, that a given set of log price changes, Δx , would have occurred. The probability is simply the product of the probabilities, $p(\Delta x; \Theta)$, for the individual observations in the data set. By taking the natural log of the probability for the whole data set, $L(\Delta x; \Theta)$, we obtain a sum of the logs of the probabilities for the individual observations. If we maximise $L(\Delta x; \Theta)$ with respect to the parameter set Θ we obtain the set of parameters which maximises the likelihood that we would have observed the set of price changes in our data set.

The density function of changes in x is given by

$$p(\Delta x; \Theta) = \sum_{j=0}^{\infty} \frac{e^{-\phi \Delta t} (\phi \Delta t)^j}{j!} n(\Delta x; \bar{\kappa} \Delta t + j \ln(1 + \bar{\kappa}), \sigma^2 \Delta t + j \gamma^2) \quad (2.19)$$

where $n(x; a, b)$ is the normal density with mean a and variance b and the vector $\Theta = (\bar{\kappa}, \sigma, \phi, \gamma)$. Assuming we have m spot price returns, $\Delta x_1, \dots, \Delta x_m$, the logarithm of the likelihood function is given by

$$\ln L(\Delta x; \Theta) = \sum_{i=1}^m \ln p(\Delta x_i, \Theta) \quad (2.23)$$

Therefore in order to estimate the parameters we simply maximise this function with respect to the parameters Θ (this can be done relatively easily using a spreadsheet solver).

Example : Maximum Likelihood Estimation of the Jump-Diffusion Parameters

The data for this example consists of daily Brent crude oil spot prices from the 20 December 1994 to the 18 February 1999 which gives 1080 data items and $\Delta t = 0.00274$.

Maximum likelihood estimation gives:

$$\bar{\kappa} = 0.0000, \sigma = 0.1938, \phi = 307.7, \gamma = 0.02057$$

$$\text{Sample Standard Deviation of returns} = 0.3451$$

A potentially undesirable empirical property of this method of estimating jump parameters is that it tends to converge on the smallest and most frequent jump component of the actual data. Energy price return time series can often be characterised as having numerous different jump components typically ranging from very high

frequency, low volatility to low frequency, high volatility jumps. We would usually want capture the lower frequency, high volatility component with a jump model. Maximum likelihood estimation may therefore not have the properties we are looking for in a jump parameter estimation procedure. Empirical analysis suggests that the Recursive Filter discussed in section 2.10 does pick out the lower frequency higher volatility jump components.

2.12 Incorporating Seasonal Patterns into the Models

Probably the most important variable after price with a strong seasonal component is volatility. There are two main ways that seasonal variations in volatility can be incorporated into the models discussed in this chapter – as a time varying pattern or as a random variable with time varying parameters. If we assume that volatility follows a predictable (i.e. non-random) seasonal pattern then this is equivalent to replacing the constant volatility parameter, σ , by a function of time, $\sigma(t)$, which repeats each seasonal cycle. We can estimate the function $\sigma(t)$ from historical data by using a rolling 30 day sample standard deviation for example⁹. Figure 2.11 illustrates the Henry Hub natural gas seasonal spot volatility over the period from 1 April 1999 to 31 March 2000 estimated using this method.

Implementing the seasonal volatility pattern into Monte Carlo simulation is straightforward – at each time step we simply apply the spot volatility appropriate to the current time point in the relevant simulation equation (e.g. equation (2.3) or (2.16)). If we are modelling volatility as random then we can incorporate the seasonal pattern into the volatility of volatility (and also the mean reversion rate and level if required) in equation (2.11).

Seasonal patterns in the mean reversion rate can also be included in a similar way. Mean reversion rates for most energy prices, with the obvious exception of electricity, are relatively low. This implies that typically at least a years worth of daily data is required to obtain reasonably accurate estimates of the mean reversion rate. In electricity markets the spot price can often be observed on an hourly or half-hourly basis¹⁰ which gives between roughly 720 and 1440 data points per month. The mean reversion rates are also much higher in electricity spot markets than other energy markets which makes estimation easier (although the spot price volatility is also higher which tends to increase the estimation error). Therefore it is possible for electricity spot prices to estimate seasonal mean reversion rates on a monthly or even weekly basis. Figure 2.12 shows estimates of the mean reversion rate for UK electricity spot prices on a weekly basis for the period from January 1999 to September 1999.

Figure 2.12 indicates that there is no strong seasonal pattern in the mean reversion rate of the UK spot electricity price over the year 1999 although there is a strong downward trend in the mean reversion rate. This can be attributed to the fact that the

⁹ In practice this “raw” estimate of the seasonal volatility function should be smoothed, for example by fitting a smooth functional form such as a polynomial (see Press *et al* (1992)).

¹⁰ For example in the UK and Australian spot electricity is settled on a half-hourly basis.

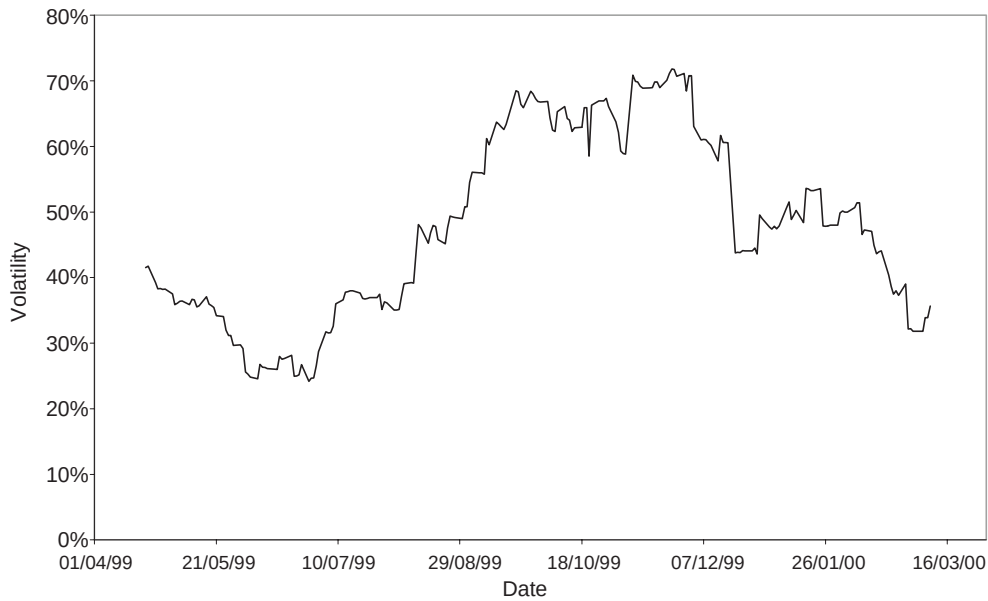


FIGURE 2.11 Henry Hub Natural Gas Seasonal Spot Volatility Over the Period from 1 April 1999 to 31 March 2000

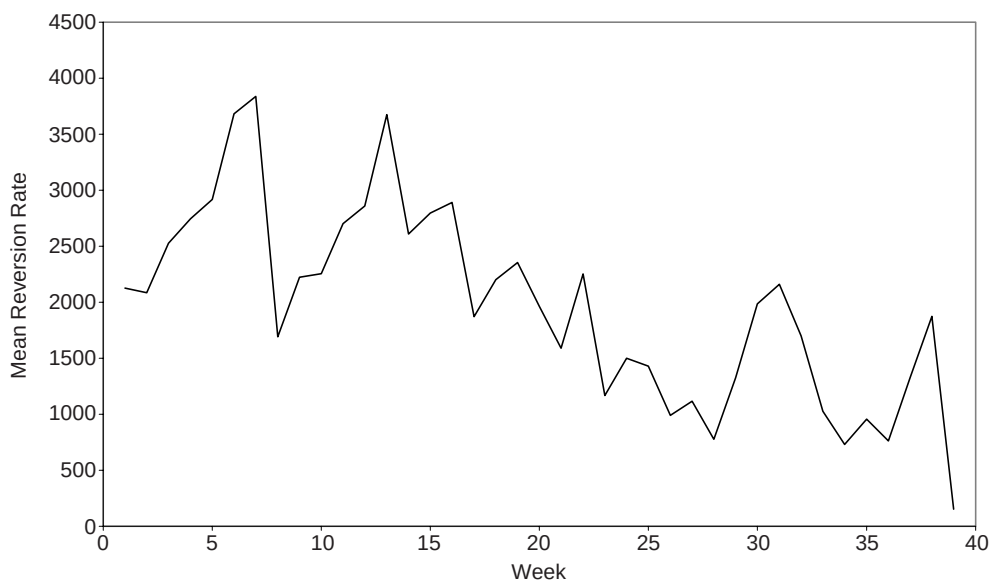


FIGURE 2.12 Mean Reversion Rate of UK Electricity Pool Prices on a Weekly Basis for January 1999 to September 1999

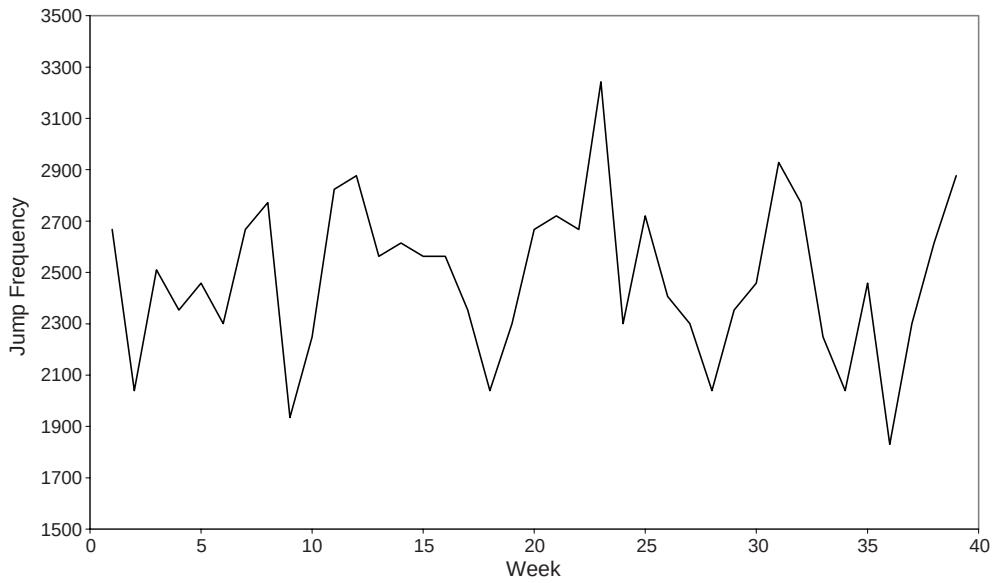


FIGURE 2.13 (a) Jump Frequency of UK Electricity Pool Prices on a Weekly Basis for January 1999 to September 1999

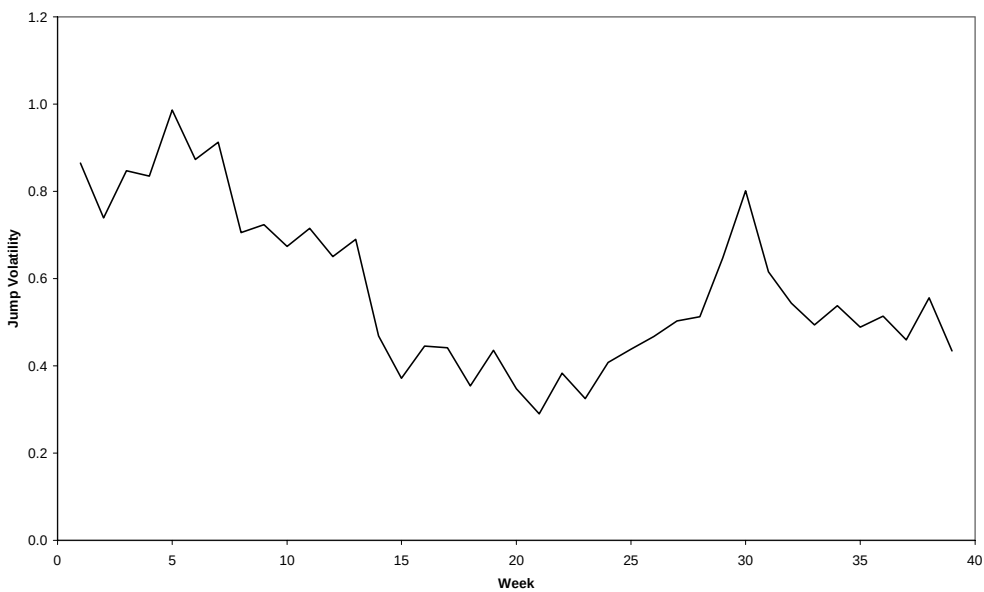


FIGURE 2.13 (b) Jump Volatility of UK Electricity Pool Prices on a Weekly Basis for January 1999 to September 1999

UK market is not yet fully competitive and the spot price behaviour is dominated by the behaviour of key players in the market.

Seasonal patterns in jump frequency and jump volatility are also important, particularly in electricity markets where demand is strongly determined by seasonal weather patterns. Regarding the estimation of seasonal jump frequency and jump volatility, similar comments as for the mean reversion rate apply. For energies other than electricity, jump frequencies and volatilities are in general relatively low and therefore a significant period of daily data (at least the order of a year) would be needed to obtain reasonable historical estimates. However, for electricity data it is possible to obtain reliable estimates on a monthly or even weekly basis. Figure 2.13 (a) and 2.13 (b) plot weekly estimates of jump frequency and jump volatility for the same UK data as figure 2.12.

The jump frequency does not show a strong seasonal pattern and appears to be roughly constant over the period of the data and should therefore probably be assumed constant. The jump volatility does show a reasonable seasonal pattern together with a similar downward trend as the mean reversion rate. The extension of Monte Carlo simulation to handle the seasonal mean reversion and jump parameters is the same as for volatility. At each time step the relevant value of the parameter for that particular point in time is obtained from the table of seasonal values and used to increment the spot price over the next time step¹¹.

An alternative method of estimation of seasonal parameter values is to imply them from traded energy option prices. This method is complicated to implement, computationally demanding and it is difficult to obtain stable estimates. We touch on these techniques in chapter 8.

2.13 Conclusions

In this chapter we have discussed how the standard model for financial market prices, the Black-Scholes-Merton GBM model, can be extended to give more realistic behaviour for energy prices. The key extensions are mean reversion, stochastic volatility and jumps. Mean reversion captures the propensity of spot prices to return to a long term level. We showed that the unpredictability of energy spot price volatility can be accounted for using a stochastic volatility model. The combination of the jump model we described together with mean reversion allowed us to reflect the sudden large changes in energy spot prices, particularly electricity. We demonstrated how all of these extensions can be related back to the original BSM model. The Monte Carlo simulation of all of the extensions was described as an essential technique for understanding and evaluating complex models and as a numerical technique for pricing energy derivatives. Finally, we explained how estimation methods for the parameters of these extended models can be implemented.

¹¹ In practice the historically estimated seasonal patterns should be smoothed in the same manner as for seasonal volatility cycles by fitting a smooth function such as a polynomial (see Press *et al* (1992)).